Unicast Routing Protocols

Contents

- Distance Vector Routing
- Link State Routing

Unicast Routing Protocols

- A host or a router has a routing table with an entry for each destination, or a combination of destinations, to route IP packets.
- A routing table can be either static or dynamic.
  - A static table is one with manual entries.
  - A dynamic table is one that is updated automatically when there is a change somewhere in the Internet.

Contd...

- A routing protocol is a combination of rules and procedures that lets routers in the Internet inform each other of changes.
- It allows routers to share whatever they know about the internet or their neighborhood.

Optimization

- A router receives a packet from a network and passes it to another network. A router is usually attached to several networks.
- When it receives a packet, to which network should it pass the packet?

The decision is based on optimization:

- Which of the available pathways is the optimum pathway?
- What is the definition of the term optimum?

Contd...

- One approach is to assign a cost for passing through a network – Cost metric.
- The metric assigned to each network depends on the type of protocol.
- Some simple protocols:
  - Routing Information Protocol (RIP)
  - Open Shortest Path First (OSPF)
  - Border Gateway Protocol (BGP)

Contd...

- **Routing Information Protocol (RIP),** treat all networks as equals.
- The cost of passing through a network is the same; it is one hop count.
- So if a packet passes through 10 networks to reach the destination, the total cost is 10 hop counts.

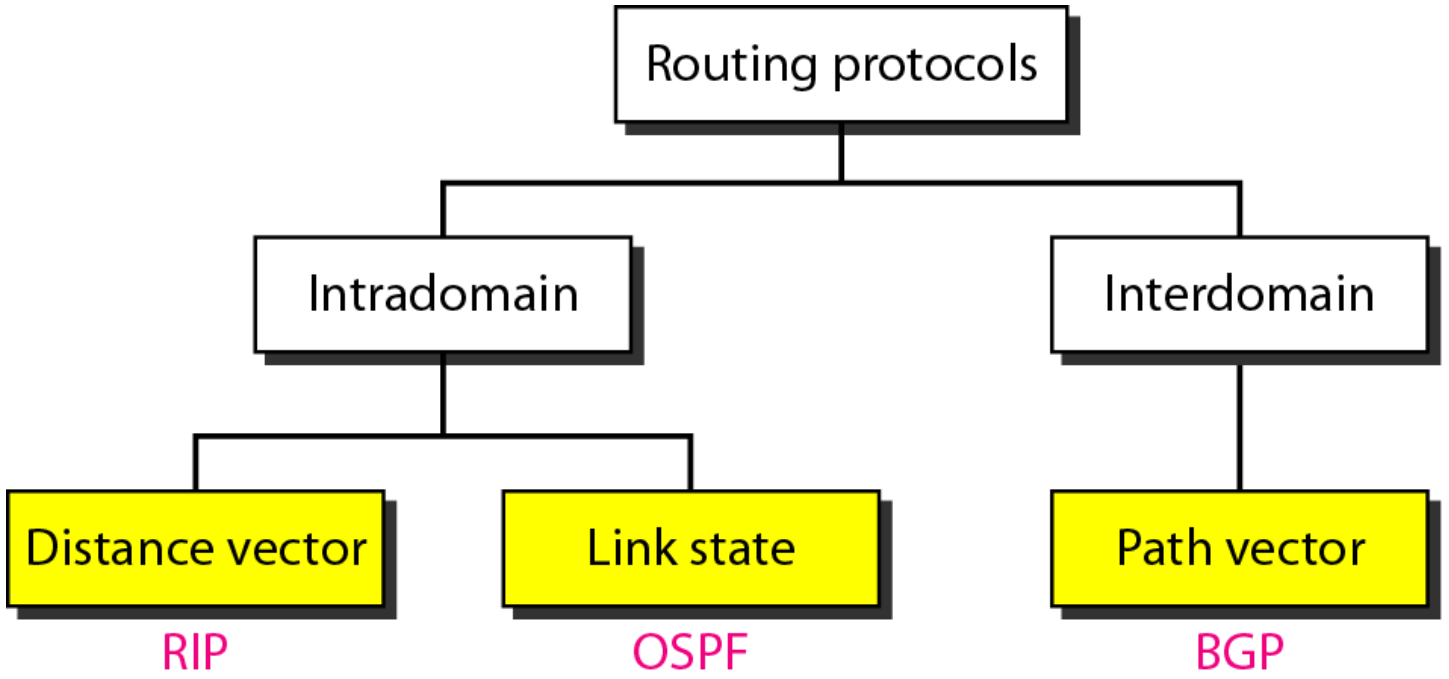
Contd...

- Open Shortest Path First (OSPF), allow the administrator to assign a cost for passing through a network based on the type of service required.
- A route through a network can have different costs (metrics). For example, if maximum throughput is the desired type of service, a satellite link has a lower metric than a fiber-optic line.
- On the other hand, if minimum delay is the desired type of service, a fiber-optic line has a lower metric than a satellite link.

Contd...

- Routers use routing tables to decide the best route.
- OSPF protocol allows each router to have several routing tables based on the required type of service.
- In the **Border Gateway Protocol (BGP),** the criterion is the policy, which can be set by the administrator.
- The policy defines what paths should be chosen.

Unicast Routing Protocols



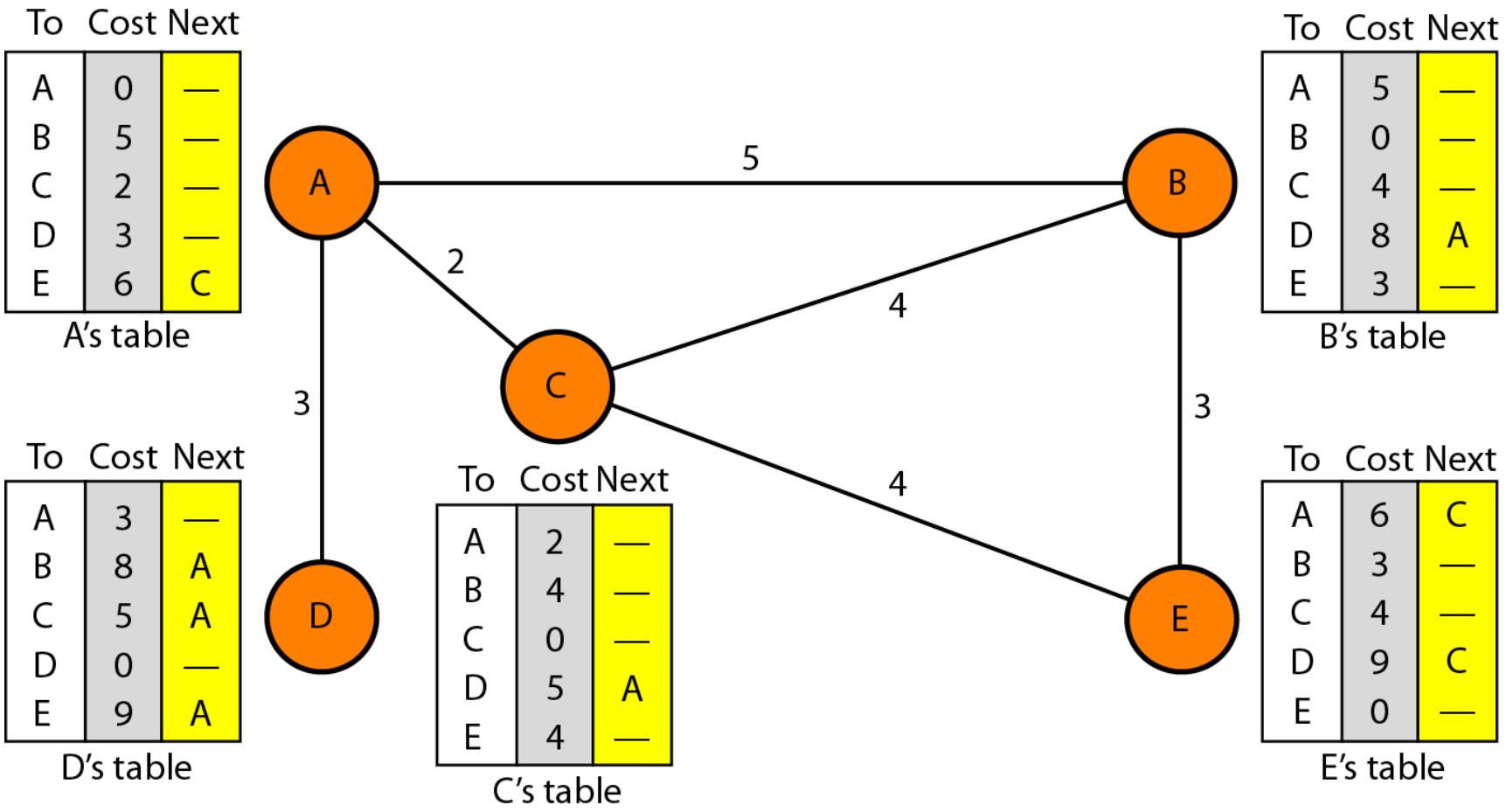
Contd...

- Routing Information Protocol (RIP) is an implementation of the distance vector protocol.
- Open Shortest Path First (OSPF) is an implementation of the link state protocol.
- Border Gateway Protocol (BGP) is an implementation of the path vector protocol.

Distance Vector Routing

- In distance vector routing, the least-cost route between any two nodes is the route with minimum distance.
- In this protocol, each node maintains a vector (table) of minimum distances to every node.
- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).

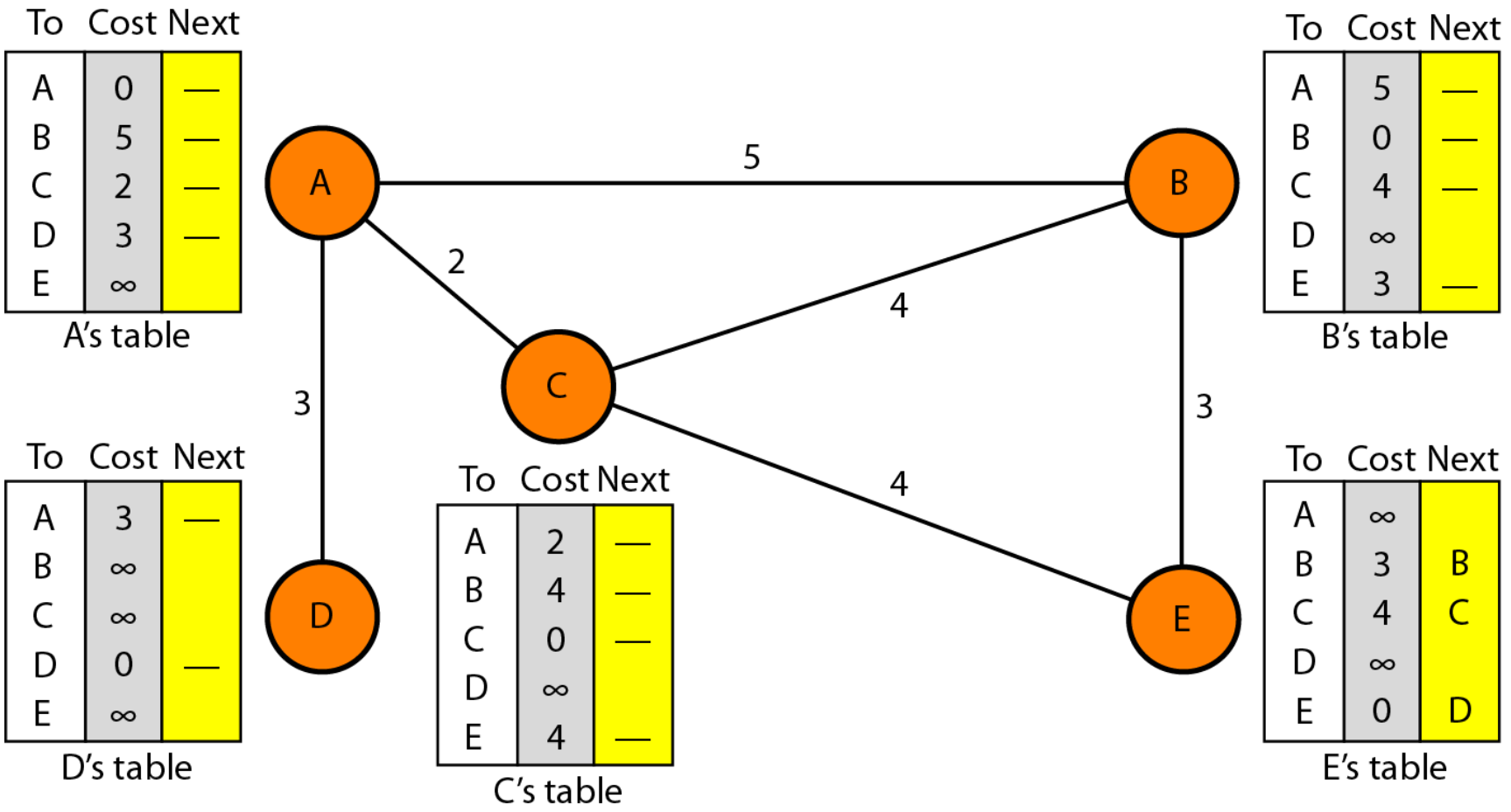
Distance vector routing tables



Initialization

- Each node knows how to reach any other node and the cost.
- At the beginning, however, this is not the case.
- Each node can know only the distance between itself and its immediate neighbors, those directly connected to it.
- The distance for any entry that is not a neighbor is marked as infinite (unreachable)

Initialization of tables in distance vector routing

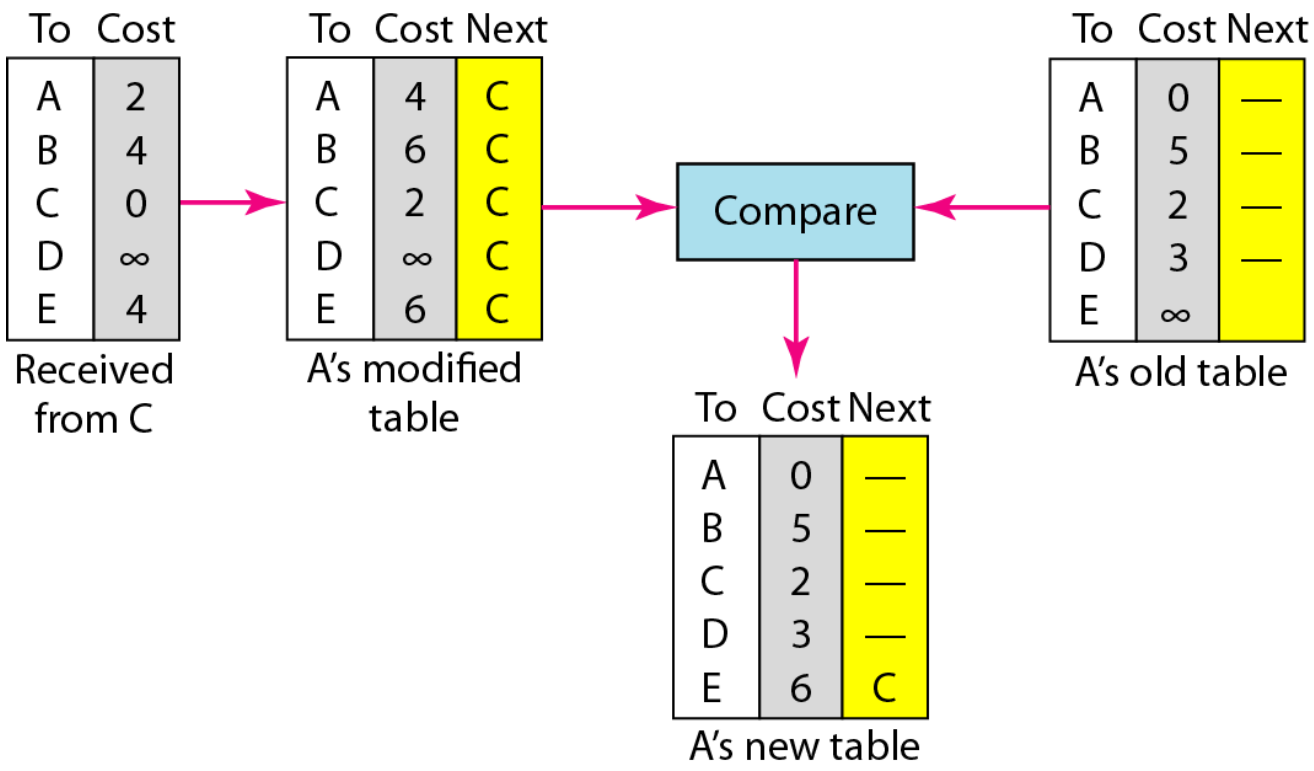


Contd...

Note

In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change.

Updating in distance vector routing



Updating

When a node receives a two-column table from a neighbor, it needs to update its routing table. Updating takes three steps:

1. The receiving node needs to add the cost between itself and the sending node to each value in the second column. The logic is clear. If node C claims that its distance to a destination is x mi, and the distance between A and C is y mi, then the distance between A and that destination, via C, is x + y mi.
2. The receiving node needs to add the name of the sending node to each row as the third column if the receiving node uses information from any row. The sending node is the next node in the route.

Contd...

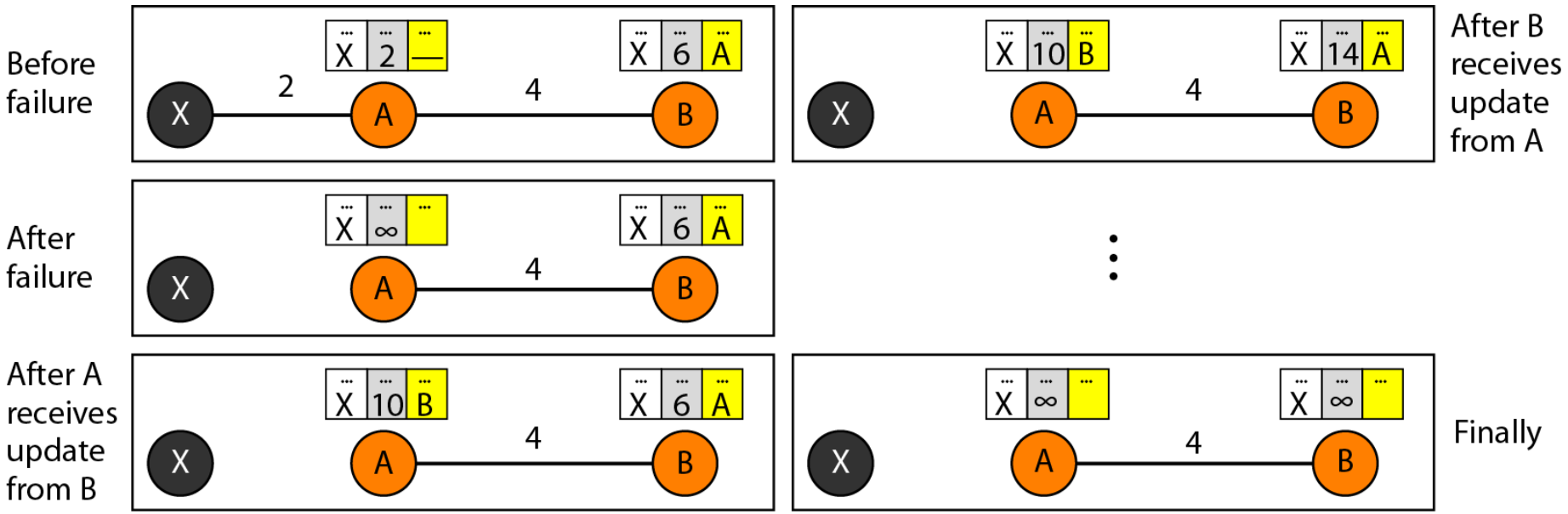
3. The receiving node needs to compare each row of its old table with the corresponding row of the modified version of the received table.

a. If the next-node entry is different, the receiving node chooses the row with the smaller cost. If there is a tie, the old one is kept.

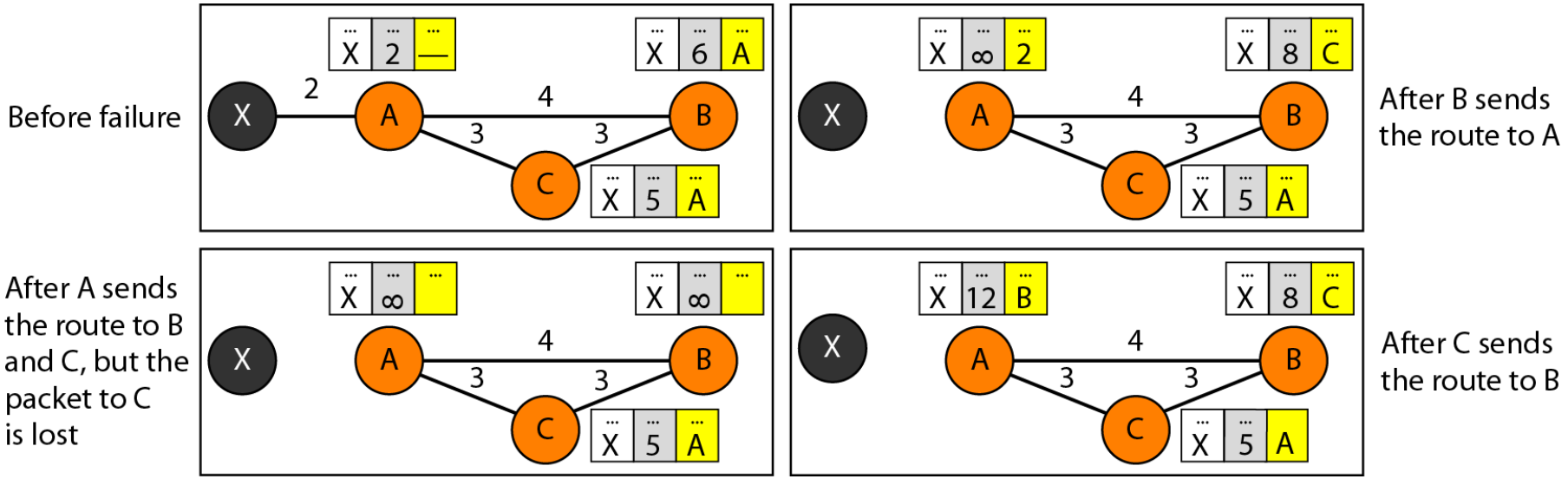
b. If the next-node entry is the same, the receiving node chooses the new row. For example, suppose node C has previously advertised a route to node X with distance 3.

- Suppose that now there is no path between C and X; node C now advertises this route with a distance of infinity. Node A must not ignore this value even though its old entry is smaller.
- The old route does not exist any more. The new route has a distance of infinity

Two-node instability



Three-node instability



Three-node instability is a problem in distance vector routing where routers have inconsistent or outdated routing information after a route failure, causing them to form a routing loop and repeatedly increase the metric toward the unreachable destination. This is known as the **count-to-infinity problem**.

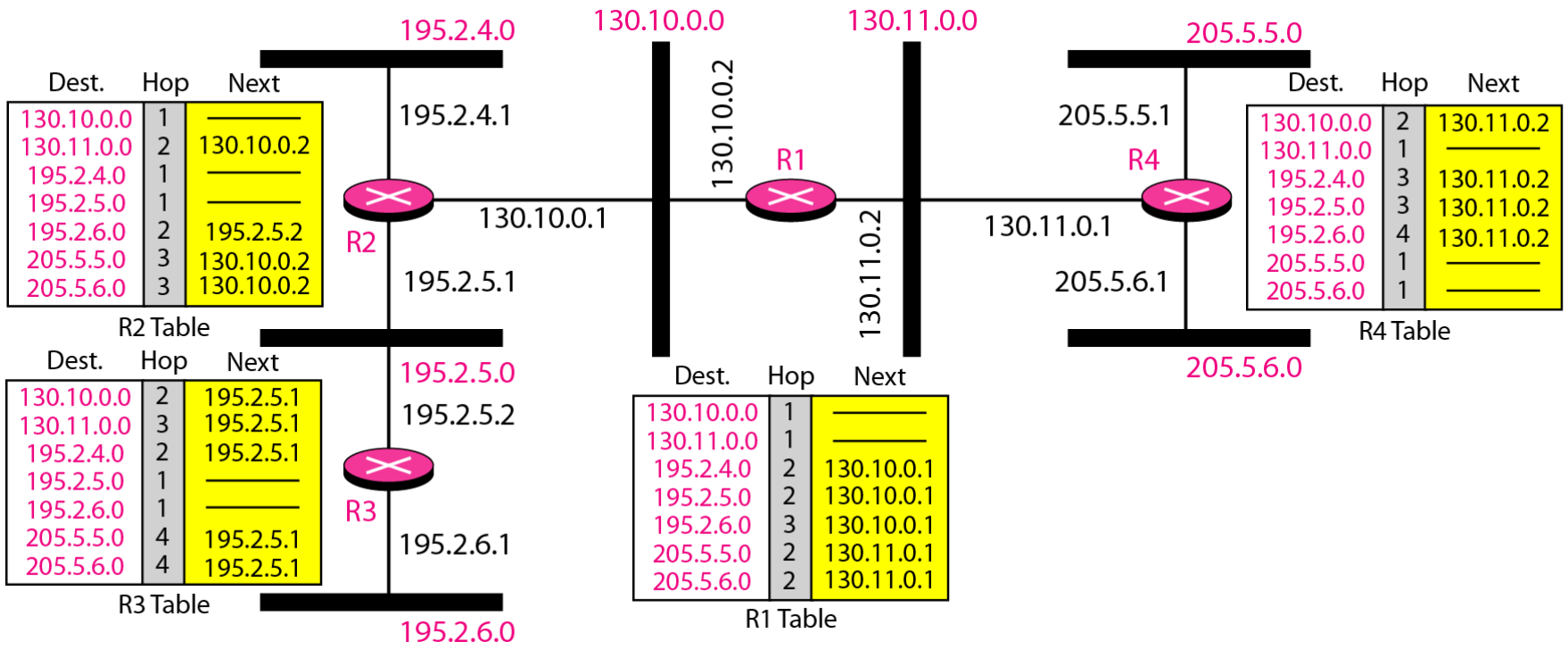
Routing Information Protocol (RIP)

- The Routing Information Protocol (RIP) is an intradomain routing protocol used inside an autonomous system.
- It is a very simple protocol based on distance vector routing.
- RIP implements distance vector routing directly with some considerations:
  - In an autonomous system, we are dealing with routers and networks (links). The routers have routing tables; networks do not.
  - The destination in a routing table is a network, which means the first column defines a network address.

Contd...

- The metric used by RIP is very simple; the distance is defined as the number of links (networks) to reach the destination. For this reason, the metric in RIP is called a hop count.
- Infinity is defined as 16, which means that any route in an autonomous system using RIP cannot have more than 15 hops.
- The next-node column defines the address of the router to which the packet is to be sent to reach its destination.

Example of a domain using RIP



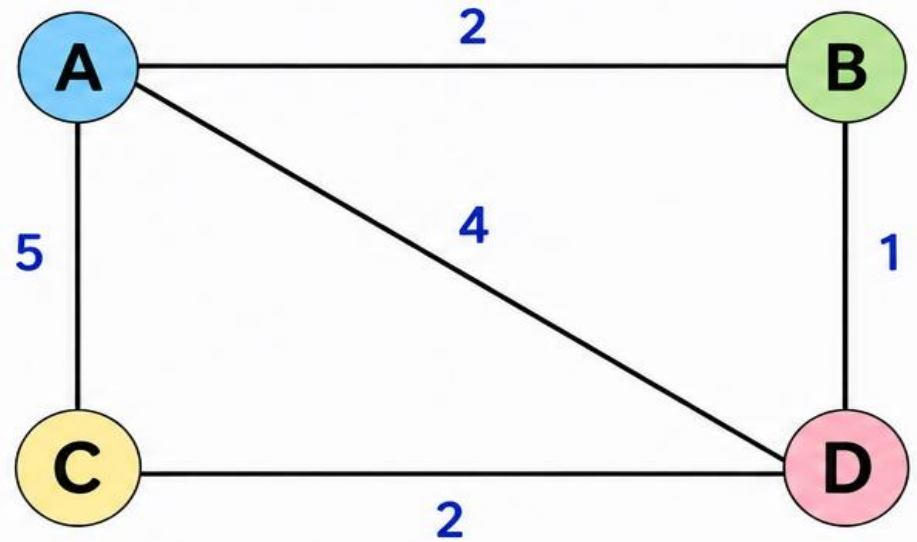
DISTANCE VECTOR ROUTING – NUMERICAL PROBLEM

PROBLEM

Consider the network shown in the figure.

1. Write the initial distance vector table at each router.
2. After exchanging their vectors once, compute the updated distance vectors.
3. Repeat the process until the algorithm converges.
4. What is the shortest path and its cost from router A to router D?

NETWORK TOPOLOGY



(Number on link = Cost)

NOTES

- ✓ Each router maintains a table with (Destination, Cost, Next Hop).
- ✓ Routers exchange their distance vectors with neighbors.
- ✓ Use Bellman–Ford equation:

$$D_x(y) = \min_{v \in N(x)} [c(x,v) + D_v(y)]$$

$D_x(y)$: cost from router x to destination y

$c(x,v)$: cost from x to its neighbor v

$D_v(y)$: cost from neighbor v to destination y

$N(x)$: set of neighbors of router x

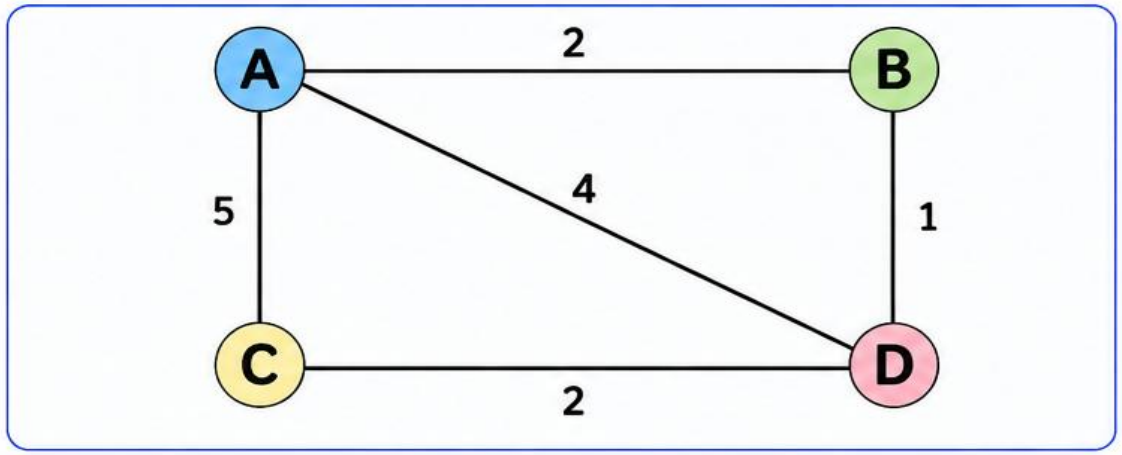
Goal: Find the shortest path and its cost from **Router A** to **Router D**.

STEP 1: INITIAL DISTANCE VECTORS (ONLY DIRECT LINKS KNOWN)

At the beginning, each router knows the cost to reach:

- Itself → cost = 0
- Its directly connected neighbors → cost of the link
- All other routers → cost = ∞ (unknown)

Next Hop: The neighbor to which the packet is sent.



LEGEND:

- 0 = Cost to self
- Link cost = Direct neighbor
- ∞ = Unknown (no information yet)
- – = Not applicable for self

Router A

Destination	Cost	Next Hop
A	0	–
B	2	B
C	5	C
D	4	D

Router B

Destination	Cost	Next Hop
A	2	A
B	0	–
C	∞	–
D	1	D

Router C

Destination	Cost	Next Hop
A	5	A
B	∞	–
C	0	–
D	2	D

Router D

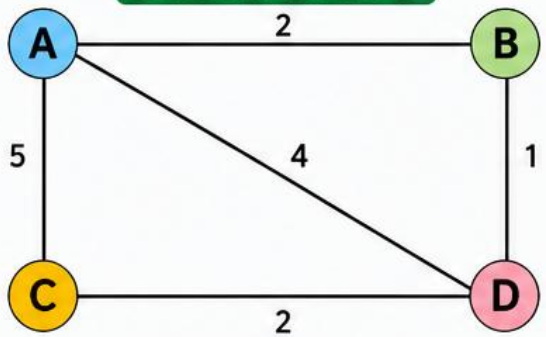
Destination	Cost	Next Hop
A	4	A
B	1	B
C	2	C
D	0	–

Summary: Each router only knows about itself and its directly connected neighbors.

STEP 2: AFTER FIRST EXCHANGE AND UPDATE

Routers exchange their distance vectors. → Each router updates its table using Bellman–Ford equation: $D_x(y) = \min_{v \in N(x)} [c(x,v) + D_v(y)]$

NETWORK TOPOLOGY



(Number on link = Cost)

HOW THE UPDATE HAPPENS (EXAMPLES)

- Router A → C via D : c(A,D) + D_D(C) = 4 + 2 = 6 (old was 5) → No update
- Router B → C via D : c(B,D) + D_D(C) = 1 + 2 = 3 (old was ∞) → Update
- Router C → A via D : c(C,D) + D_D(A) = 2 + 4 = 6 (old was 5) → No update
- Router C → B via D : c(C,D) + D_D(B) = 2 + 1 = 3 (old was ∞) → Update
- Router D → A via B : c(D,B) + D_B(A) = 1 + 2 = 3 (old was 4) → Update

LEGEND

0	= Cost to self
Green	= Direct neighbor
∞	= Infinity (unknown)
–	= Not applicable (self)

Comparison Rule

Update only if
New cost < Old cost

Router A

Destination	Cost	Next Hop
A	0	–
B	2	B
C	5	C
D	4	D

Changes for Router A:
No change in any entry.

Router B

Destination	Cost	Next Hop
A	2	A
B	0	–
C	3	D
D	1	D

Changes for Router B:
C : ∞ → 3 via D (Updated)
(A and D unchanged)

Router C

Destination	Cost	Next Hop
A	5	A
B	3	D
C	0	–
D	2	D

Changes for Router C:
B : ∞ → 3 via D (Updated)
A and D unchanged

Router D

Destination	Cost	Next Hop
A	3	B
B	1	B
C	2	C
D	0	–

Changes for Router D:
A : 4 → 3 via B (Updated)
(B, C and D unchanged)

Summary:

- After the first exchange, some entries are updated (shown in red).
- Next round of exchanges will further update the tables until no more changes occur (convergence).

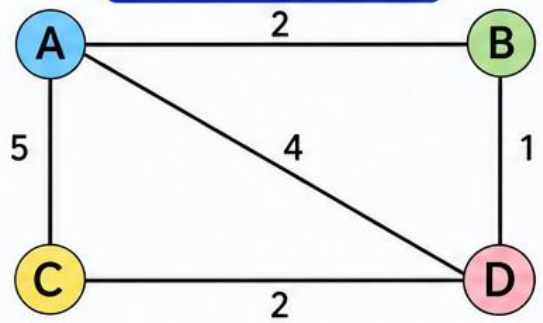
Key Takeaway

A route is updated only when the new cost is smaller than the old cost.

STEP 3: AFTER FURTHER EXCHANGES (CONVERGED – NO MORE CHANGES)

✅ **No entries change in this round.** Hence, algorithm has converged. These are the final shortest costs.

NETWORK TOPOLOGY



(Number on link = Cost)

WHAT HAPPENS IN THIS STEP?

- Routers exchange their updated vectors again.
- Each router checks all possible routes using **Bellman–Ford equation:**

$$D_x(y) = \min_{v \in N(x)} [c(x,v) + D_v(y)]$$

- No router finds a better (smaller) cost for any destination.
- **No changes in any table → Converged.**

LEGEND

0	= Cost to self
Green	= Direct neighbor
∞	= Infinity (unknown)
—	= Not applicable (self)

Router A

Destination	Cost	Next Hop
A	0	–
B	2	B
C	5	C
D	3	B

Router B

Destination	Cost	Next Hop
A	2	A
B	0	–
C	3	D
D	1	D

Router C

Destination	Cost	Next Hop
A	5	A
B	3	D
C	0	–
D	2	D

Router D

Destination	Cost	Next Hop
A	3	B
B	1	B
C	2	C
D	0	–

No changes for Router A
(Shortest routes confirmed)
A→B cost 2 via B
A→C cost 5 via C (or B–D–C also 5)
A→D cost 3 via B

No changes for Router B
(Shortest routes confirmed)
B→A cost 2 via A
B→C cost 3 via D
B→D cost 1 via D

No changes for Router C
(Shortest routes confirmed)
C→A cost 5 via A (or D–B–A also 5)
C→B cost 3 via D
C→D cost 2 via D

No changes for Router D
(Shortest routes confirmed)
D→A cost 3 via B
D→B cost 1 via B
D→C cost 2 via C

⭐ **SUMMARY:** All routing tables are stable.
No further updates occur. Algorithm has converged.

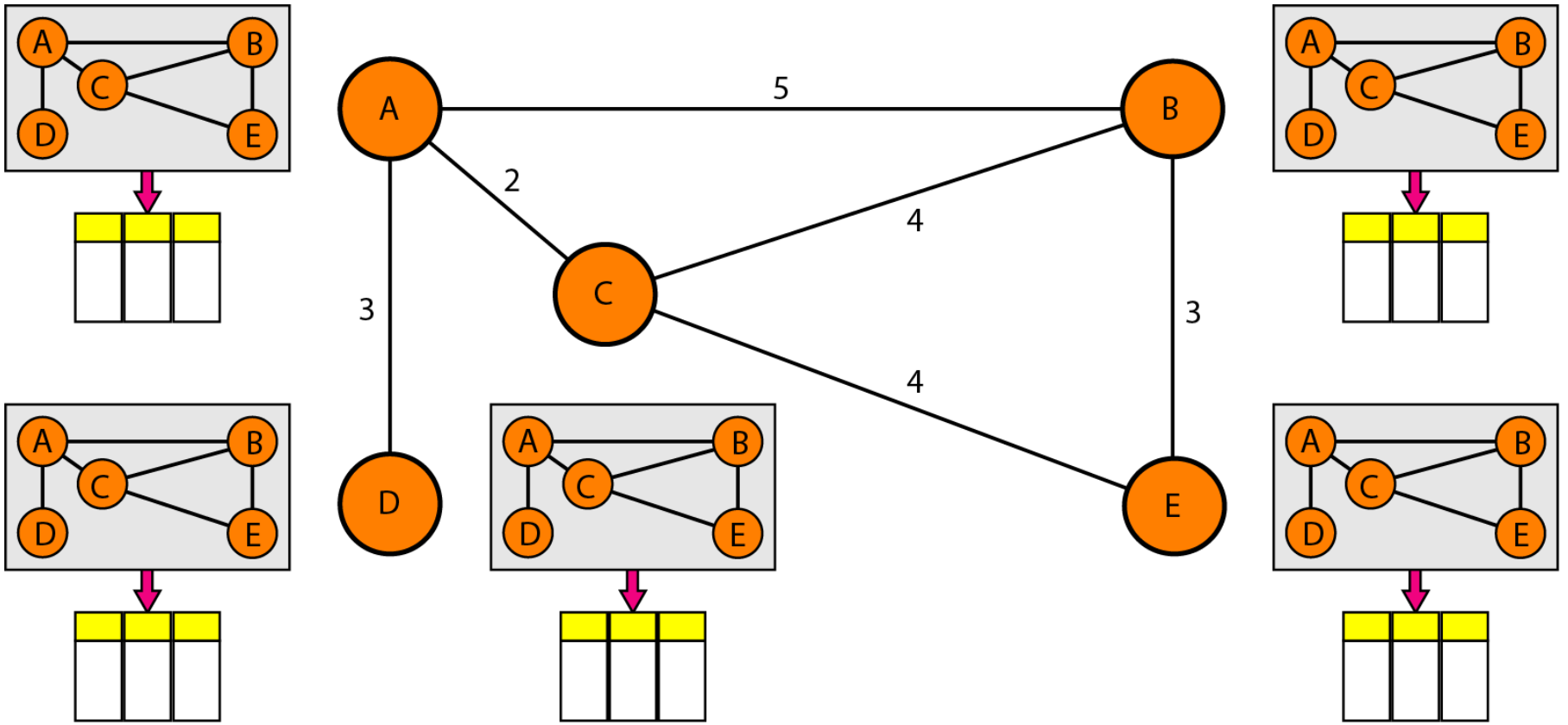
FINAL SHORTEST PATH FROMATO D
A → B → D
Cost = 2 (A–B) + 1 (B–D) = 3

Link State Routing

Link State Routing

- In link state routing, if each node in the domain has the entire topology of the domain the list of nodes and links
- How they are connected including the type, cost (metric) condition of the links (up or down)-the node can use Dijkstra's algorithm to build a routing table.

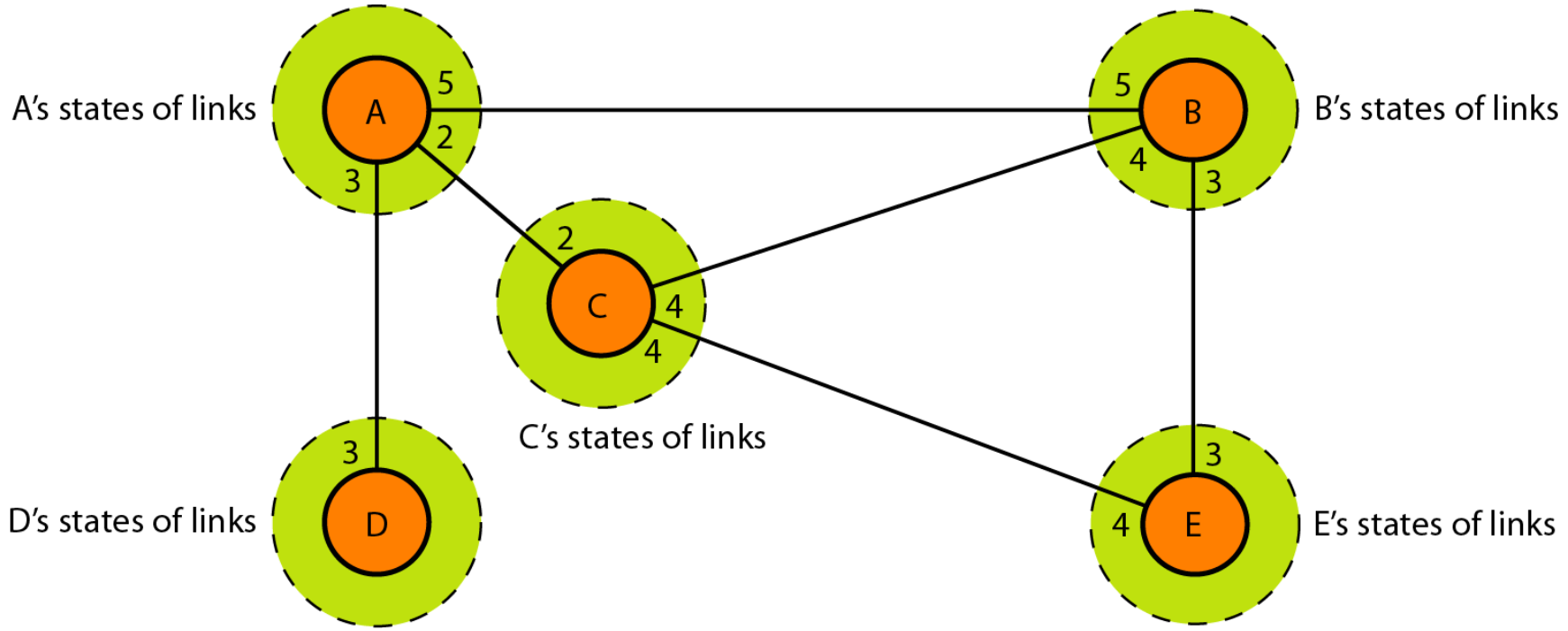
Concept of link state routing



Contd...

- Link state routing is based on the assumption that, although the global knowledge about the topology is not clear, each node has partial knowledge: it knows the state (type, condition, and cost) of its links.
- The whole topology can be compiled from the partial knowledge of each node

Link state knowledge



Building Routing Tables

- In link state routing, four sets of actions are required to ensure that each node has the routing table showing the least-cost node to every other node
  - Creation of the states of the links by each node, called the link state packet (LSP).
  - Dissemination of LSPs to every other router, called flooding, in an efficient and reliable way.
  - Formation of a shortest path tree for each node.
  - Calculation of a routing table based on the shortest path tree.

Creation of Link State Packet (LSP)

- A link state packet can carry a large amount of information.
- It carries a minimum amount link of data: the node identity, the list of links, a sequence number, and age.
- The first two, node identity and the list of links, are needed to make the topology.
- The third, sequence number, facilitates flooding and distinguishes new LSPs from old ones.
- The fourth, age, prevents old LSPs from remaining in the domain for a long time.

Contd...

LSPs are generated on two occasions:

- When there is a change in the topology of the domain. Triggering of LSP dissemination is the main way of quickly informing any node in the domain to update its topology.
- On a periodic basis. The period in this case is much longer compared to distance vector routing. As a matter of fact, there is no actual need for this type of LSP dissemination. It is done to ensure that old information is removed from the domain.

Contd...

- The timer set for periodic dissemination is normally in the range of 60 min or 2 h based on the implementation.
- A longer period ensures that flooding does not create too much traffic on the network.
- Flooding of LSPs After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbors. The process is called **flooding.**

Contd...

- The creating node sends a copy of the **LSP out of each interface**.
- A node that receives an LSP compares it with the copy it may already have. If the newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP. If it is newer, the node does the following:

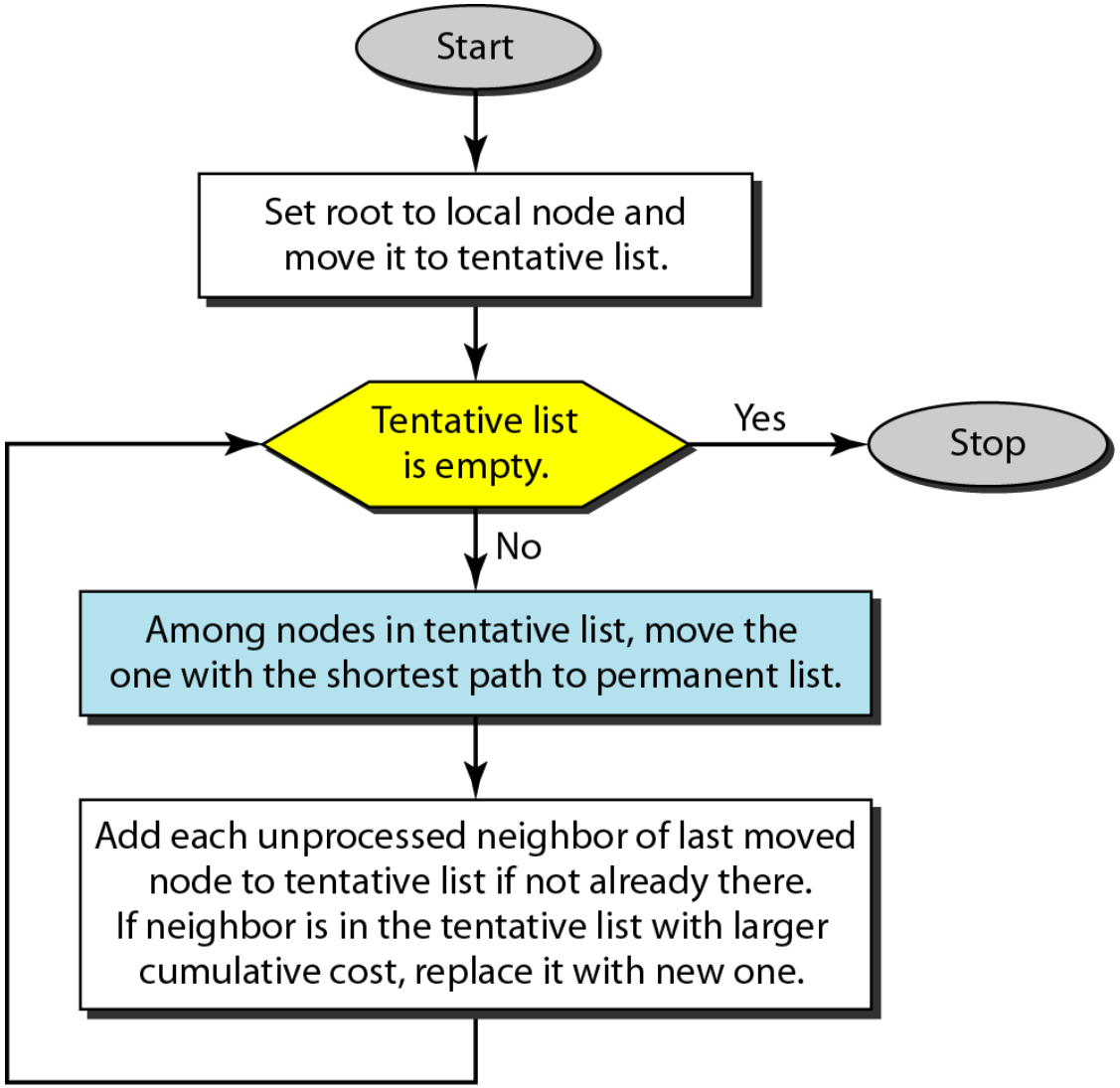
  a. It discards the old LSP and keeps the new one.

  b. It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain (where a node has only one interface)

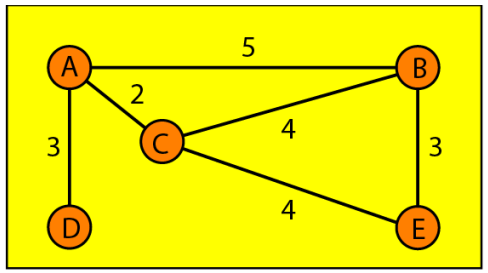
Formation of Shortest Path Tree: Dijkstra Algorithm

- The Dijkstra algorithm creates a shortest path tree from a graph.
- The algorithm divides the nodes into two sets: tentative and permanent.
- It finds the neighbors of a current node, makes them tentative, examines them, and if they pass the criteria, makes them permanent.

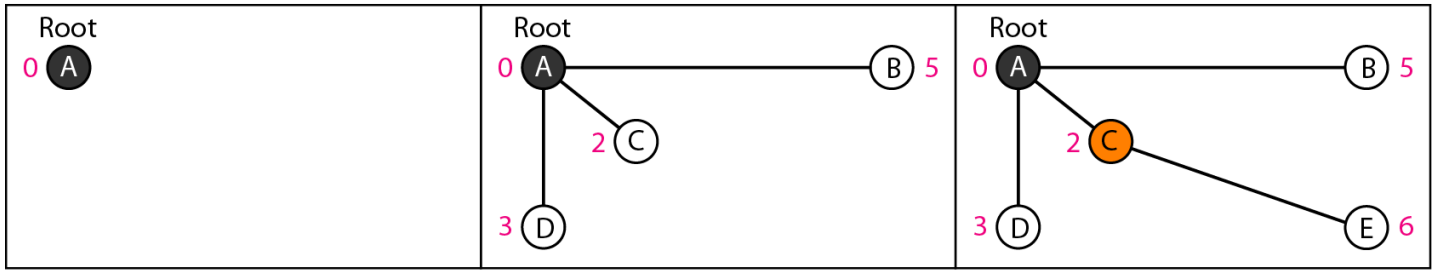
Dijkstra algorithm



Example of formation of shortest path tree



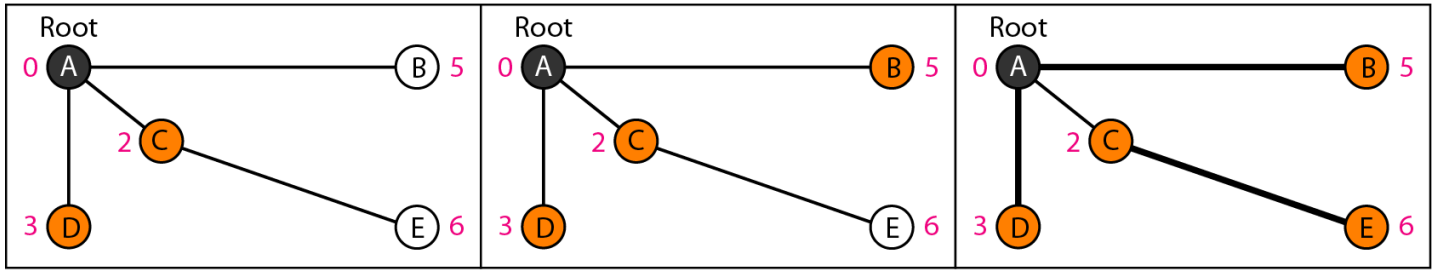
Topology



1. Set root to A and move A to tentative list.

2. Move A to permanent list and add B, C, and D to tentative list.

3. Move C to permanent and add E to tentative list.



4. Move D to permanent list.

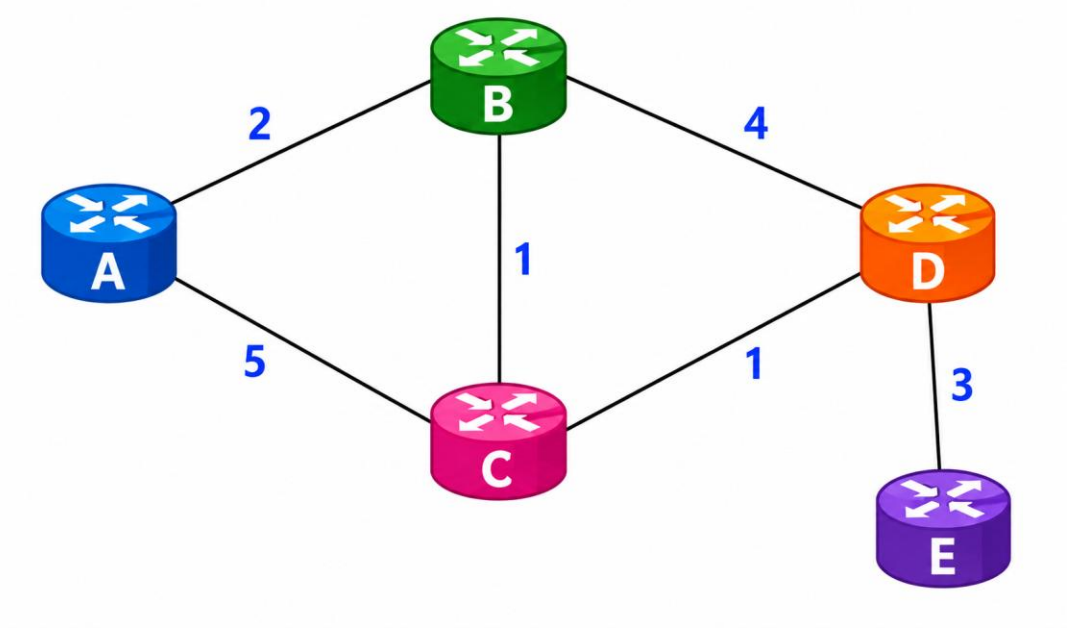
5. Move B to permanent list.

6. Move E to permanent list (tentative list is empty).

Table 22.2

Node	*Cost*	*Next Router*
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

Link State Routing example

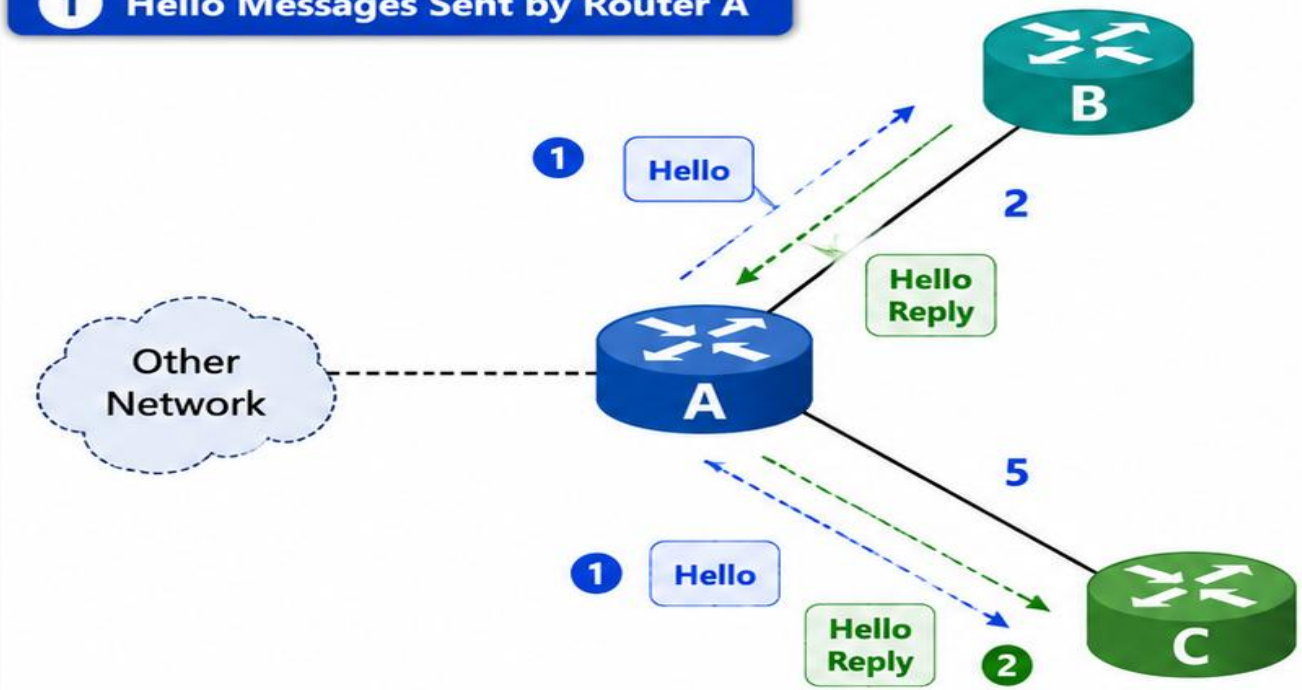


STEP 1 Discover Neighbors

Router A discovers its directly connected neighbors by sending **Hello** messages.

Link State Routing

1 Hello Messages Sent by Router A



What happens in Step 1?

1. Router A sends Hello messages out of all its active interfaces.
2. Routers B and C receive the Hello messages and send Hello Reply messages back to A.
3. When A receives the replies, it confirms that B and C are its directly connected neighbors.

Result of Step 1

Router A knows who its neighbors are.

Neighbor Table at Router A after Step 1

Neighbor	Interface	Link Cost	Status
B	Interface to B	2	Neighbor Discovered
C	Interface to C	5	Neighbor Discovered

Legend

- Hello (from A)
- Hello Reply (to A)

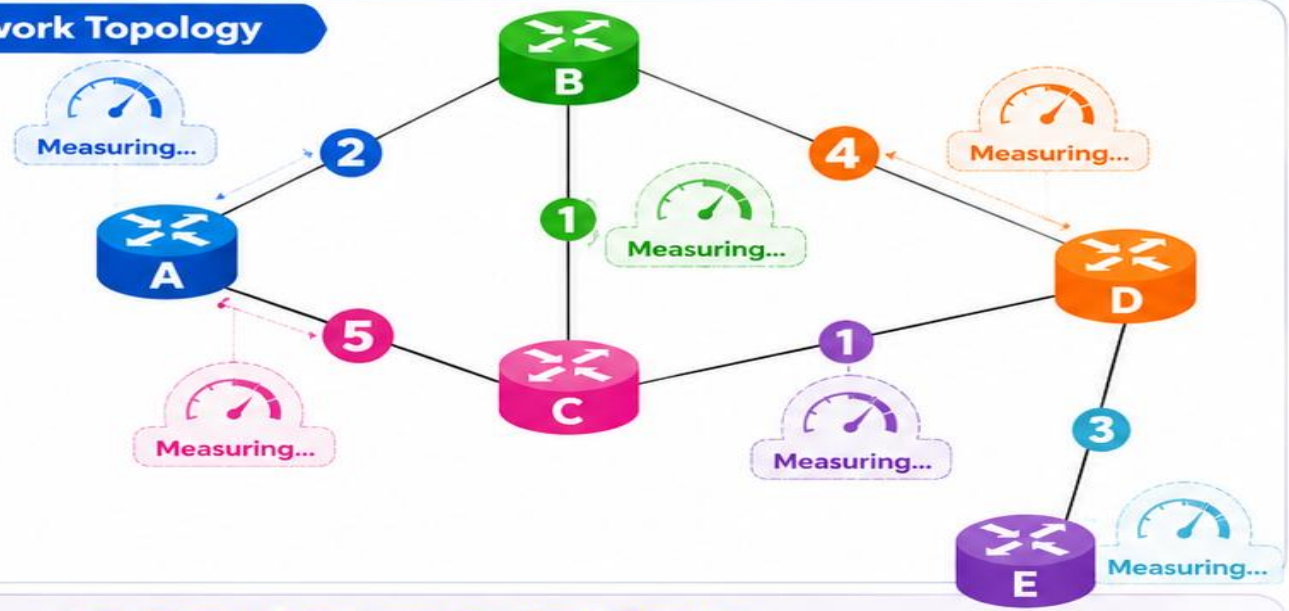
Numbers on links = Link Cost

STEP 2 Measure Link Costs

Each router measures the cost of its directly connected links.

Link State Routing

Network Topology



What happens in Step 2?

1. Each router actively measures **the cost (metric) of every link to its neighbors.**
2. The cost can be based on **bandwidth, delay, reliability,** or a combination of factors.
3. These measured costs will be included in the **Link State Packet (LSP)** in the next step.

Link Costs Measured by Each Router

Router	Neighbor	Link Cost (Metric)	Meaning
A	B	2	Cost from A to B
A	C	5	Cost from A to C
B	A	2	Cost from B to A
B	C	1	Cost from B to C
B	D	4	Cost from B to D
C	A	5	Cost from C to A
C	B	1	Cost from C to B
C	D	1	Cost from C to D
D	B	4	Cost from D to B
D	C	1	Cost from D to C
D	E	3	Cost from D to E
E	D	3	Cost from E to D

Key Takeaway

After Step 2, every router knows the cost of all its directly connected links.

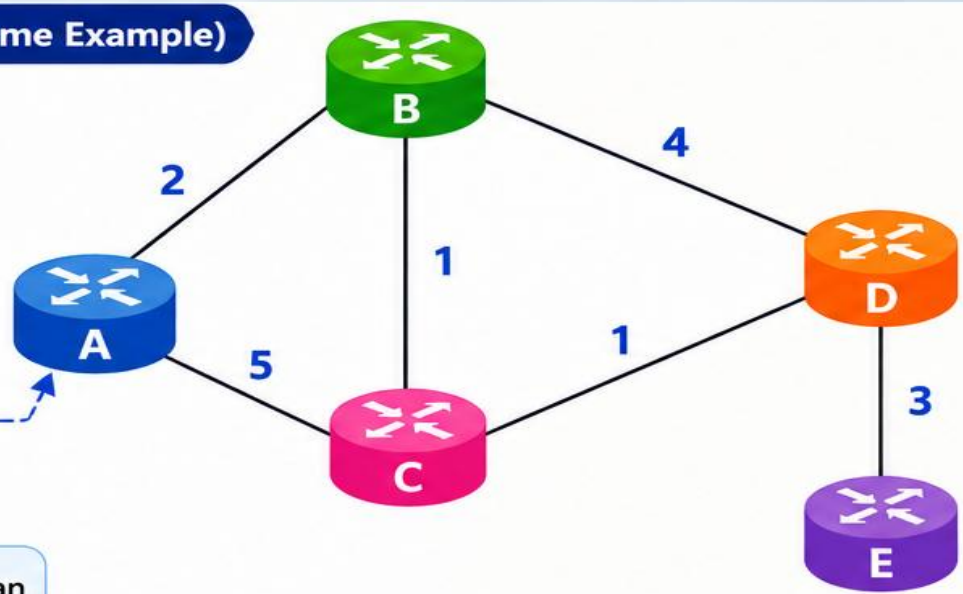
Step 2 is all about discovering **how "expensive" each link is to reach.**

STEP 3 Create Link State Packet (LSP)

Link State Routing

Each router creates an LSP that describes its directly connected links.

Network Topology (Same Example)



Router A

Neighbors and Link Costs

Neighbor	Cost
B	2
C	5

Router A will create an **LSP** containing this information.

A's Link State Packet (LSP)

Router ID	Sequence Number	Age (Seconds)	Number of Links	Link Data (Neighbor, Cost)	
				Neighbor	Cost
A	25	0	2	B	2
				C	5

This packet represents **A's local view** of the network.

What happens in Step 3?

1. Each router collects the list of its directly connected neighbors and the cost to each neighbor.
2. The router creates a **Link State Packet (LSP)** containing:
   - Router ID
   - Sequence Number (for freshness)
   - Age (time since created)
   - Number of links
   - List of (Neighbor, Cost) pairs
3. The LSP is ready to be flooded to the rest of the network in the next step.

Key Point

An LSP advertises **link information**, not the distance to all destinations.

Step 3 Output: Every router now has an LSP that describes only its own links — not the entire network.

STEP 3 Create Link State Packet (LSP) & Send to Neighbors

Link State Routing

Router A creates an **LSP** describing its directly connected links and sends it to all neighbors.

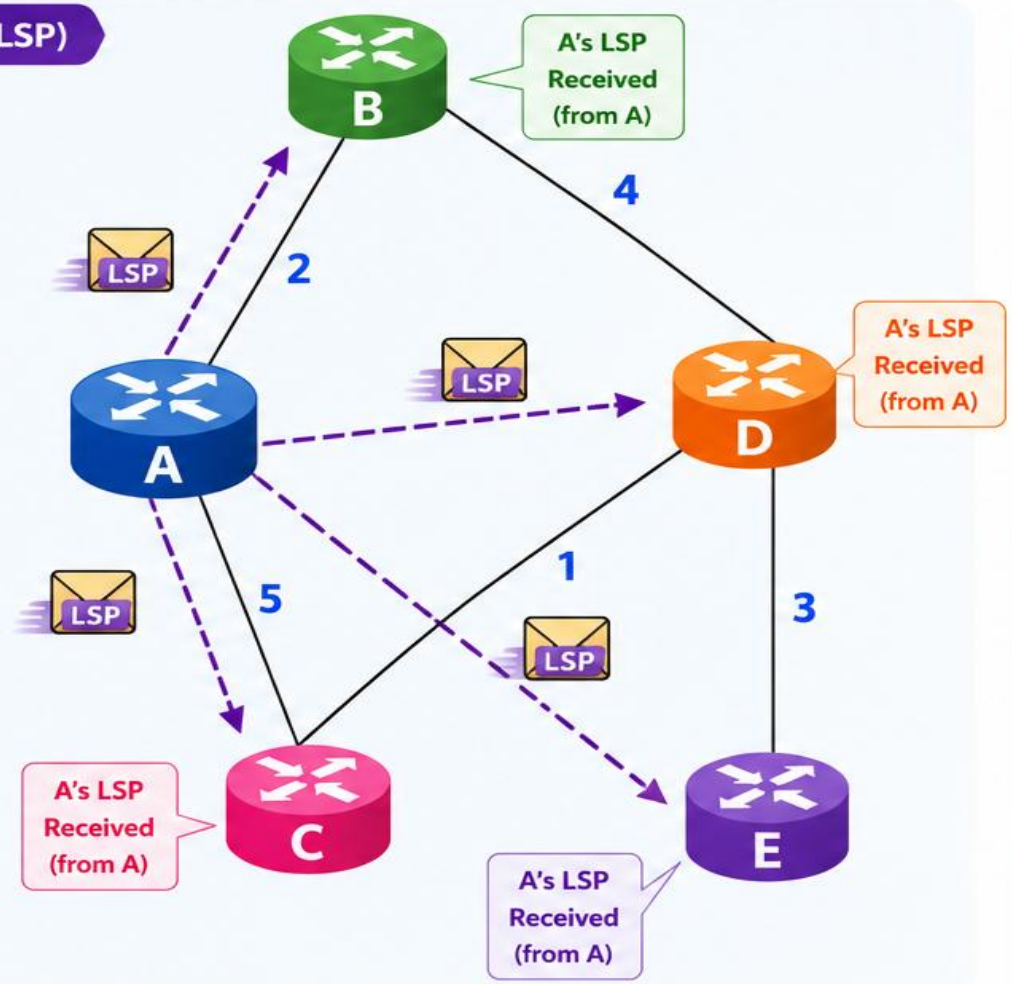
1) A Creates Its Link State Packet (LSP)

A's LSP Contents

- Router ID : A
- Sequence No : 25
- Age : 0 sec
- Number of Links : 2
- Link Data (Neighbor, Cost)

Neighbor	Cost
B	2
C	5

This LSP represents A's **local view** of the network (only its direct links).



2) A Sends (Floods) the LSP

1. Router A sends its **LSP** out of all its active interfaces.
2. Neighbors B, C, D, and E receive the **LSP** from A.
3. Each neighbor stores this **LSP** in its Link State Database (**LSDB**) (the same LSP will be flooded further in the next step).

Key Point

An LSP advertises **link information** (connected neighbors and costs), not the shortest paths.

Step 3 Summary

Router A creates an **LSP** describing its links (A–B cost 2, A–C cost 5) and sends this LSP to all neighbors **B**, **C**, **D**, and **E**.

LSP Packet sent by A

LSP Link State Packet

STEP 4 Build Link State Database (LSDB)

Link State Routing

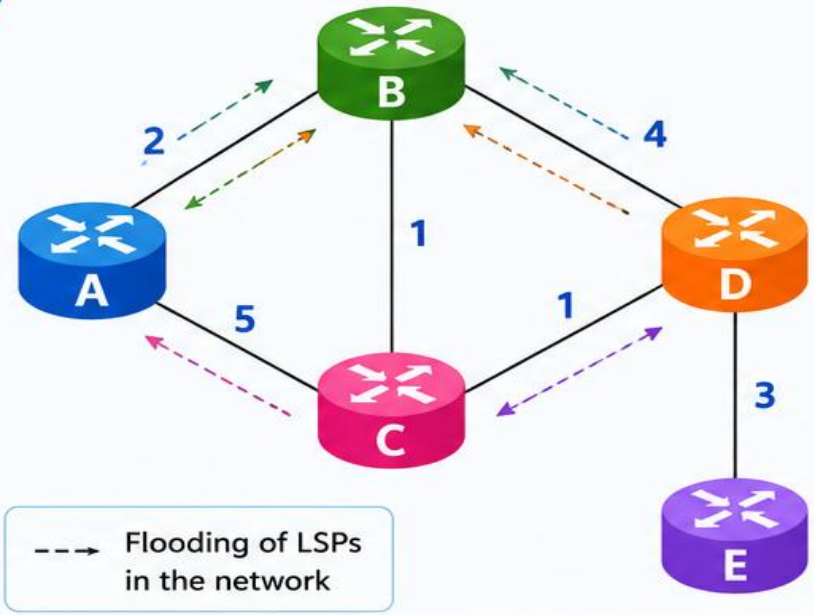
Each router receives LSPs from all other routers and builds a complete link state database.

1) All Routers Receive LSPs

After flooding, every router receives the Link State Packets (LSPs) generated by all routers in the network.

LSPs Received at Router A

- LSP from A
- LSP from B
- LSP from C
- LSP from D
- LSP from E



2) Build Link State Database (LSDB)

Router A stores the link state information from all routers in its Link State Database (LSDB).

LSDB at Router A

Advertising Router	Link (To)	Cost	Sequence No.	Age (sec)
A	B	2	25	10
	C	5	25	10
B	A	2	32	8
	C	1	32	8
	D	4	32	8
C	A	5	18	12
	B	1	18	12
	D	1	18	12
D	B	4	20	9
	C	1	20	9
	E	3	20	9
E	D	3	15	11

Every router will have (almost) the same LSDB.

3) What Does LSDB Contain?

- **Advertising Router ID** (Who created the LSP)
- **Links to Neighbors** (Which neighbors are connected)
- **Link Costs (Metrics)** (Cost to reach each neighbor)
- **Sequence Number** (For freshness of information)
- **Age / Lifetime** (Time since the LSP was created)

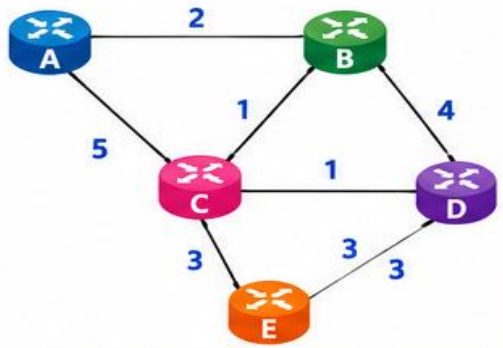
Step 4 Outcome: Router A now has a complete and consistent view of the entire network topology in its Link State Database (LSDB). This database is the basis for the next step (running Dijkstra's algorithm).

STEP 6 Run Dijkstra's Algorithm (SPF)

Each router runs SPF (Dijkstra) on its LSDB to find the shortest paths from itself to all other routers.

Link State Routing

Our Network



Router A's LSDB (from Step 4)

Link (From)	Link (To)	Cost
A	B	2
A	C	5
B	A	2
B	C	1
B	D	4
C	A	5
C	B	1
C	D	1
C	E	3
D	B	4
D	C	1
D	E	3
E	D	3

Dijkstra's Algorithm at Router A

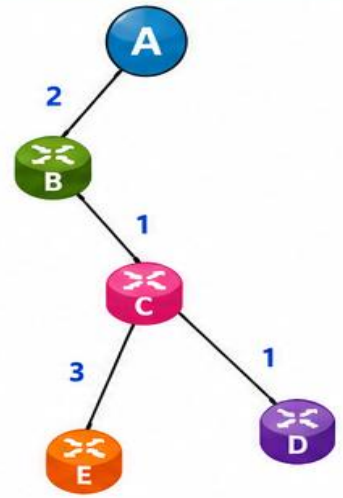
Step	Nodes in Shortest Path Tree (Permanent Set)	Tentative Distance from A				Predecessor (Next Hop from A)	
		B	C	D	E		
0 Initialization	{ A }	2 (A)	5 (A)	∞ (-)	∞ (-)	B: A C: A D: - E: -	Start from A. A is permanent (cost 0). Initialize neighbors.
1 Select B (min cost = 2)	{ A, B }	–	3 (B)	6 (B)	∞ (-)	B: A C: B D: B E: -	Pick B (cost 2). Update via B: A→B→C = 2+1=3 (better than 5) A→B→D = 2+4=6
2 Select C (min cost = 3)	{ A, B, C }	–	–	4 (C)	6 (C)	B: A C: B D: C E: C	Pick C (cost 3). Update via C: A→B→C→D = 3+1=4 (better than 6) A→B→C→E = 3+3=6
3 Select D (min cost = 4)	{ A, B, C, D }	–	–	–	6 (C)	B: A C: B D: C E: C	Pick D (cost 4). Update via D: A→B→C→D→E = 4+3=7 (6 is better, so no change)
4 Select E (min cost = 6)	{ A, B, C, D, E }	–	–	–	–	B: A C: B D: C E: C	Pick E (cost 6). All nodes are permanent. **Algorithm complete.**

Final Shortest Path Costs from A

A	B	C	D	E
0	2	3	4	6

These costs represent the minimum total cost from A to every router.

Shortest Path Tree (Rooted at A)



Router A's Routing Table (Output)

Destination	Shortest Path (from A)	Cost	Next Hop
A	A	0	-
B	A → B	2	B
C	A → B → C	3	B
D	A → B → C → D	4	B
E	A → B → C → E	6	B

Step 6 Result: Router A now knows the **shortest path** and **next hop** to reach every destination in the network. It installs this routing table in its routing table and uses it to forward packets.

Open Shortest Path First

- The OSPF protocol is an intradomain routing protocol based on link state routing. Its domain is also an autonomous system.
- Areas To handle routing efficiently and in a timely manner, OSPF divides an autonomous system into areas.
- An area is a collection of networks, hosts, and routers all contained within an autonomous system.
- An autonomous system can be divided into many different areas. All networks inside an area must be connected.
- Routers inside an area flood the area with routing information.

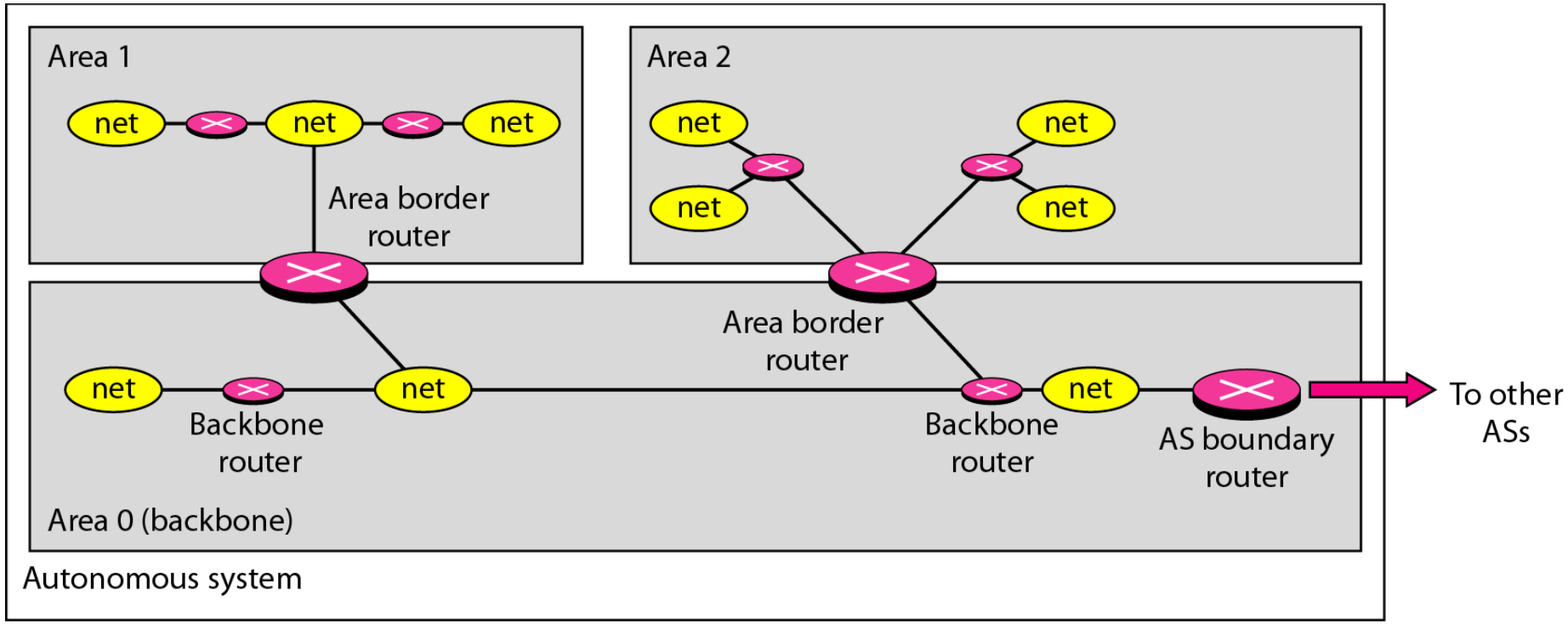
Contd...

- At the border of an area, special routers called area border routers summarize the information about the area and send it to other areas.
- Among the areas inside an autonomous system is a special area called the backbone; all the areas inside an autonomous system must be connected to the backbone.
- The backbone serves as a primary area and the other areas as secondary areas.
- This does not mean that the routers within areas cannot be connected to each other. The routers inside the backbone are called the backbone routers.

Contd...

- A backbone router can also be an area border router.
- If, because of some problem, the connectivity between a backbone and an area is broken, a virtual link between routers must be created by an administrator to allow continuity of the functions of the backbone as the primary area.
- Each area has an area identification. The area identification of the backbone is zero.

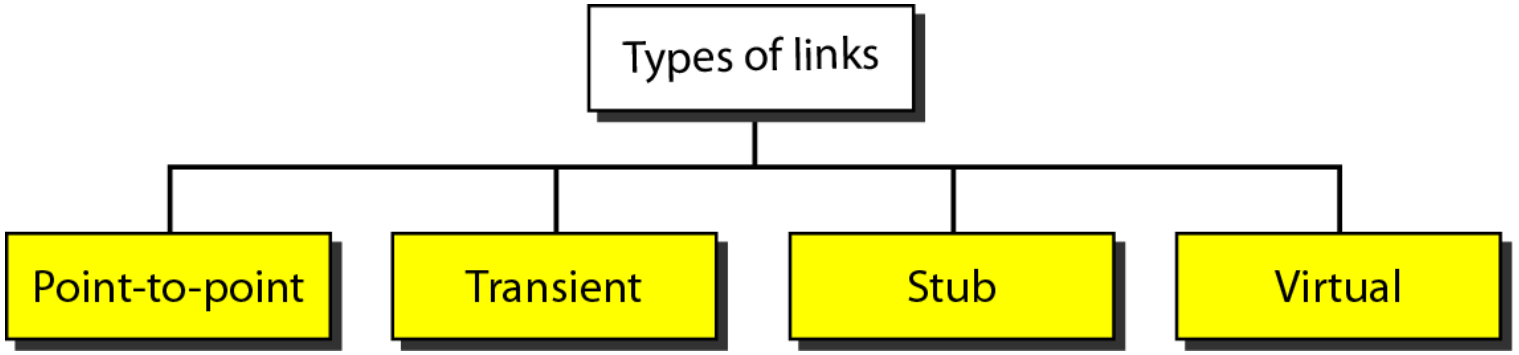
Areas in an autonomous system



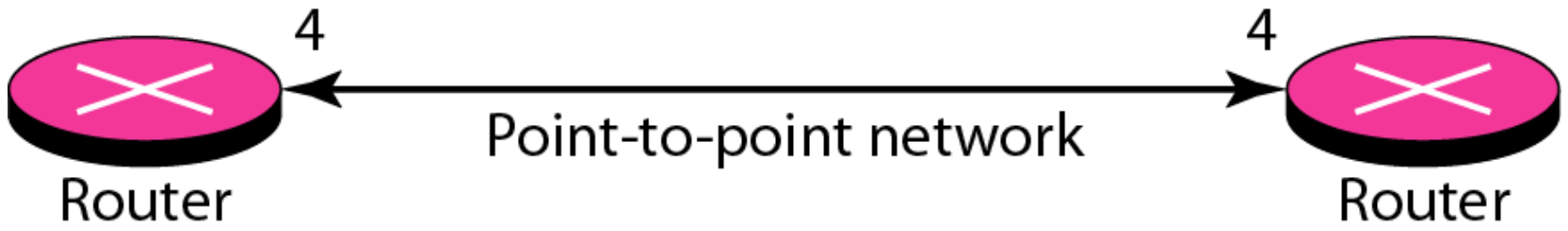
Metric

- The OSPF protocol allows the administrator to assign a cost, called the metric, to each route.
- The metric can be based on a type of service (minimum delay, maximum throughput, and so on).
- As a matter of fact, a router can have multiple routing tables, each based on a different type of service.

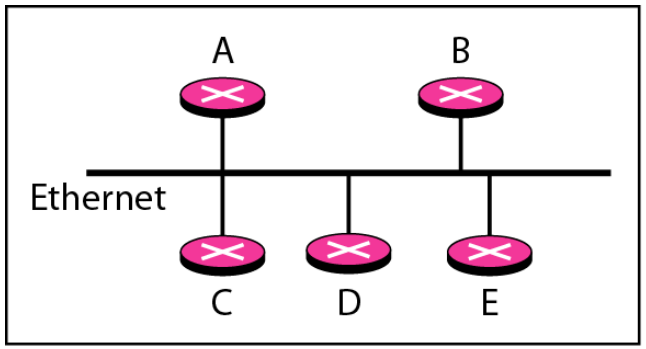
Types of links



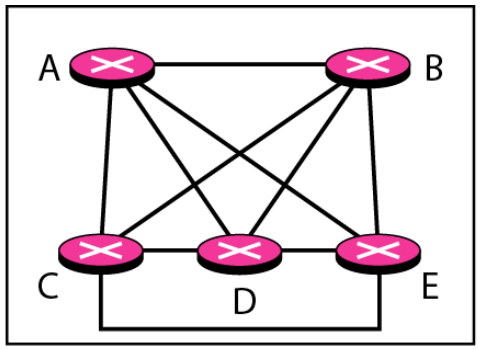
Point-to-point link



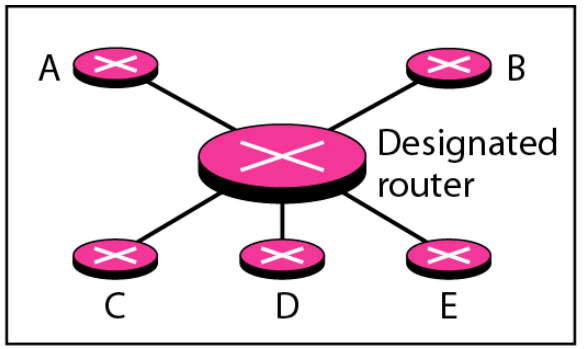
Transient link



a. Transient network

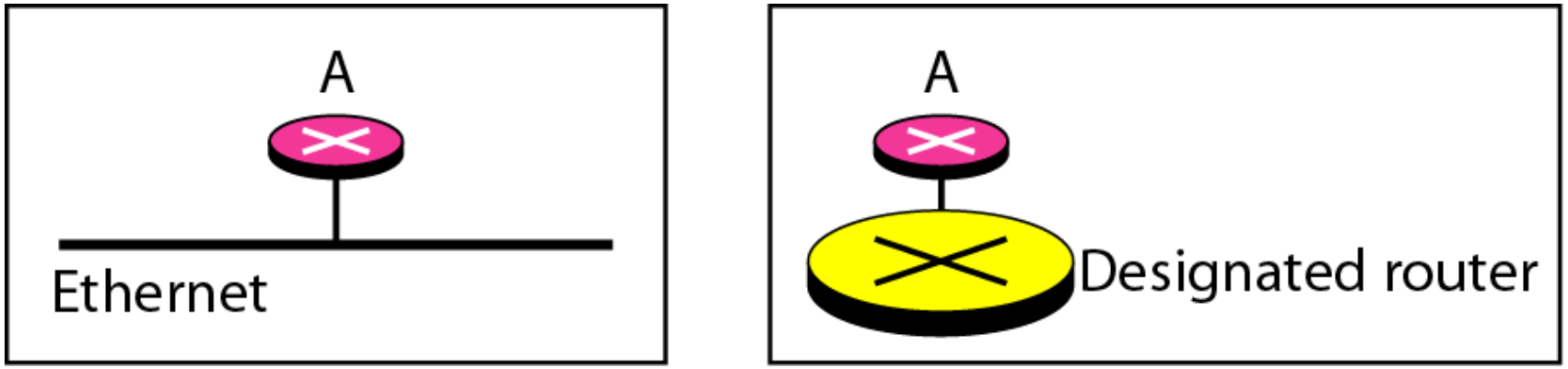


b. Unrealistic representation



c. Realistic representation

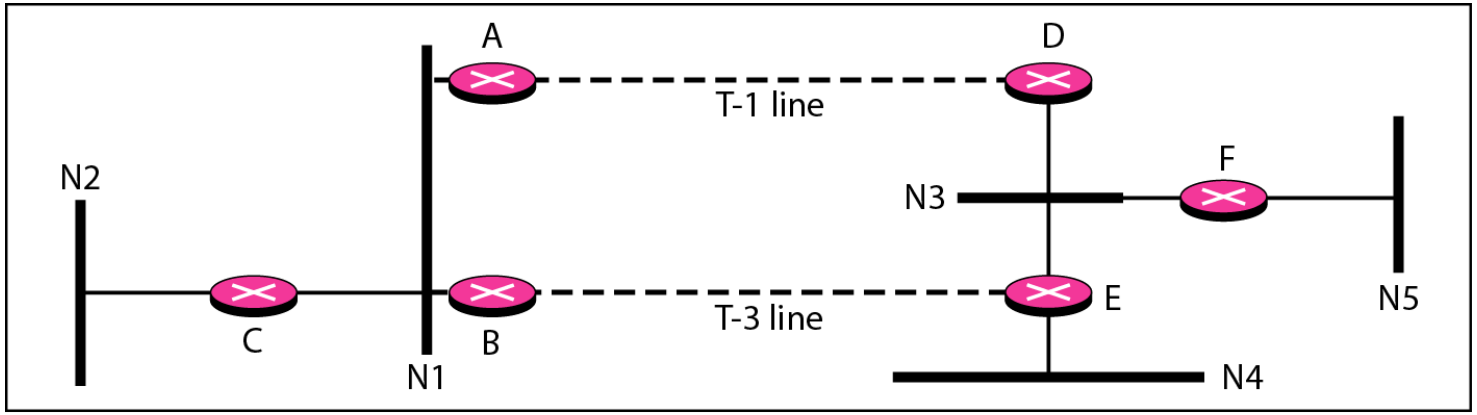
Stub link



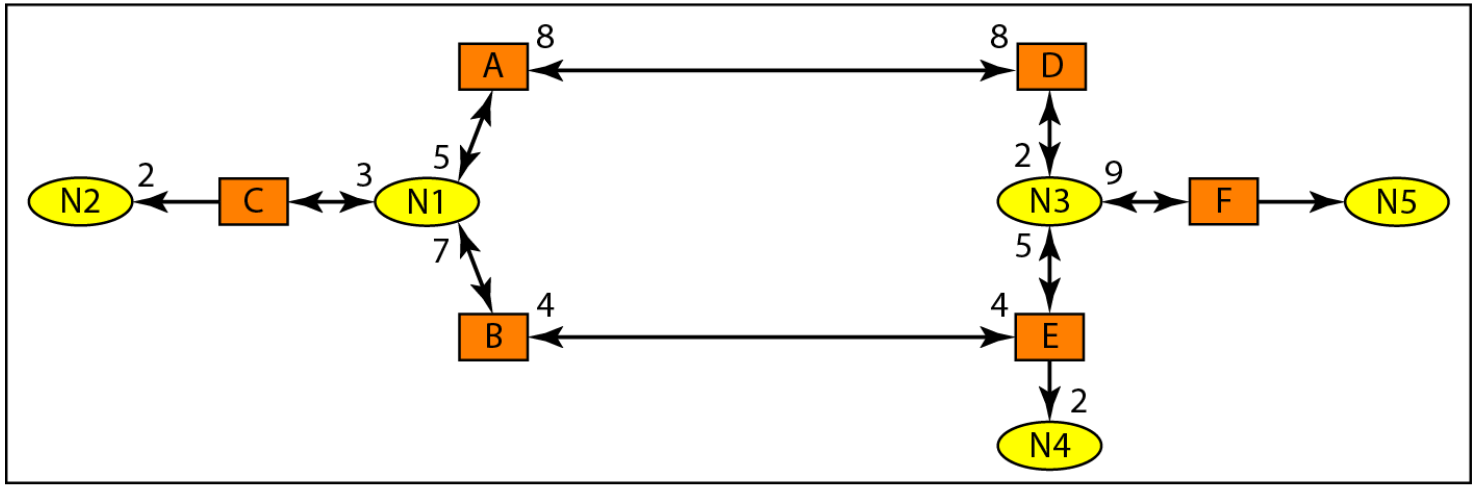
a. Stub network

b. Representation

Example of an AS and its graphical representation in OSPF



a. Autonomous system



b. Graphical representation

ROUTING ALGORITHMS

BELLMAN–FORD ALGORITHM

Used in Distance Vector Routing (e.g., RIP)

How it works (Simple Steps)

1 Share



Each router sends its distance (cost) table to its neighbors.

2 Calculate



Router calculates new cost to each destination using info from neighbors.

3 Update



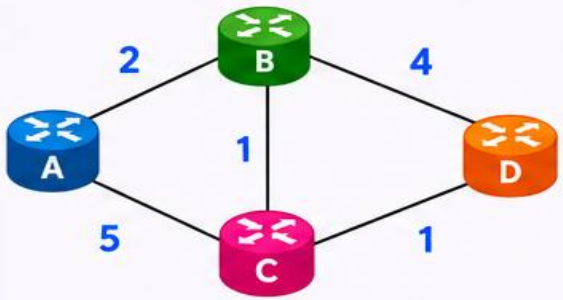
Update the table with the smallest cost.

4 Repeat



Repeat until tables stop changing.

Example Network



Distance Vector Table at A (Example)

Destination	Cost	Next Hop
A	0	–
B	2	B
C	3	B
D	4	B

Algorithm Used

BF

Bellman–Ford Algorithm

Finds the shortest path by relaxing all routes repeatedly based on neighbors' information.

DIJKSTRA'S SHORTEST PATH FIRST (SPF)

Used in Link State Routing (e.g., OSPF)

How it works (Simple Steps)

1 Discover



Find all neighbors and link costs.

2 Create LSP



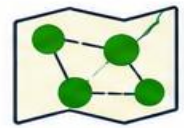
Create Link State Packet about links.

3 Fiood



Send LSP to all routers in the network.

4 Build Map



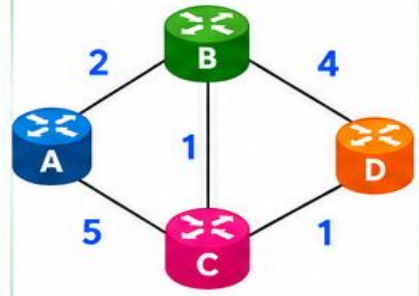
All routers build the same complete network map.

5 Run SPF

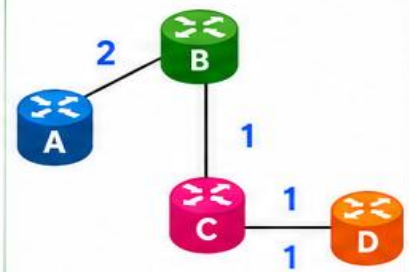


Run Dijkstra (SPF) to find shortest paths.

Example Network



Shortest Path Tree from A (using Dijkstra)



Routing Table at A

Destination	Cost	Next Hop
A	0	–
B	2	B
C	3	B
D	4	B

Algorithm Used

DJ

Dijkstra's Shortest Path First (SPF)

Finds the shortest path from the source to all nodes using the complete network map.

Link State Vs Distance Vector

Link State

- Link states algorithm is an algorithm **using global information**
- Each node **talks with all other nodes**, but tell them only the cost of it's directly comparison of some of their attribute
- **Message complexity**: With link state, every node has to keep the information about the cost of each link within the network.
- Many times, if any of the link cost is changed, all the nodes are **updated.**

Distance Vector

- The distance vector algorithm is **iterative, asynchronous, and distributed**
- each node **talks to only its directly connected neighbors**, but provides its neighbor with least cost estimates from itself to all the nodes.
- **Message complexity**: with distance vector algorithm, message is exchanged between two hosts which are directly connected to each other.
- Change of cost in the link which is belong to the least cost path for one of the nodes, the DV algorithm will update the new value. But if the change doesn't belong to the least cost part between 2 hosts, there will **no updating**.

Contd...

Link State

- **Speed of convergence**: can converge faster in comparison of later.
- Such probability is less.
- **Robustness:** For LS, when a router is down, it can broadcast a wrong cost for the closest one. LS node is computing for its own forwarding table and other node do the calculation for themselves. **Better than DV.**

Distance Vector

- **Speed of convergence**: can converge slowly and have routing loops while the algorithm is converging.
- DV algorithm also suffers from the **count to infinity** problem.
- **Robustness:** DV, the wrong least cost path can be passed to more than one or all of the node so the wrong calculation will be process in the entire net work. This problem of DV is much **worse than LS algorithm**.

Activity

Laboratory Exercise: 7

Configure Basic wireless networking using Cisco Packet Tracer

https://www.youtube.com/watch?v=Jp0hhYpNSYY